BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes. P2 pp.118-129

Defensive Design, Validation and Testing

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: Answer all four before turning to the answers.

1. Define a variable.

2. Give the first index of an array.

3. State what output means in a program.

4. For array [4, 8, 2], update index 1 to 5 and state the sum of indices 0 and 1.

Today's objectives

Explain defensive design and input validation. Distinguish validation from authentication.

Key vocabulary - Read, cover, recall

defensive design
Designing programs to handle invalid input and misuse.

validation
Checking entered data against defined rules.

authentication
Checking a user's identity.

boundary data
Valid data at the extreme limits of an allowed range.

Write one definition in your own words:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes. P2 pre-118129

Learn and practise

New learning - Four key ideas

DEFENSIVE DESIGN	Anticipate unsuitable input; handle it safely.
VALIDATION	Check data against a type, range or length rule.
AUTHENTICATION	Check an identity. This is not validation.
TESTING	Try normal, boundary, invalid and erroneous data.

MARK VALIDATION - ACCEPT 0 TO 100 INCLUSIVE

-1	0	56	100	101
Reject	Accept	Accept	Accept	Reject

'hello' is erroneous (wrong type); it must be handled safely.

I Do - Worked example

A program accepts a mark from 0 to 100 inclusive. Choose test data and explain the expected results.

1. The valid mark range is 0 to 100 inclusive.
2. Normal: 56 -> accept. Boundaries: 0 and 100 -> accept.
3. Invalid numeric values: -1 and 101 -> reject.
4. Erroneous type: 'hello' -> reject or handle safely.

We Do - Guided practice

Classify 100 and 101 for a 0-100 mark rule.

BLACK PEN - complete your explanation:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes. Page 18 of 29

Think and write

Code First - Predict > Write > Test > Improve

Read the OCR Exam Reference Language code before testing it.

```
value = 260
if value >= 0 AND value <= 255 then
print("Fits")
else
print("Out of range")
endif
```

1. Predict whether 260 fits in unsigned 8-bit binary.

2. Change value to 255 and predict the result.

3. Explain why the allowed range is 0-255.

You Do - Independent exam questions

Q1. Define input validation. [2 marks]

Q2. Give two boundary values and one invalid value for marks from 0 to 100 inclusive. [3 marks]

Q3. Distinguish authentication and validation; explain why iterative testing is useful. [4 marks]

Exit Ticket - True or False?

Answer before checking Page 4.

1. Validation and authentication mean exactly the same thing. ☐ T ☐ F

2. Validation checks data against a rule. ☐ T ☐ F

3. An invalid mark of 101 must be accepted in the range 0-100. ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes. R2 pre-118-129

Review & improve

GREEN PEN: Mark fairly. Tick correct points and fix any mistakes.
Use the mark allocations; then rewrite one answer in full.

Do Now - Answers

Q1 Named storage for a program value. ☐

Q2 0. ☐

Q3 Information produced/displayed by the program. ☐

Q4 New array [4, 5, 2]; sum = 4 + 5 = 9. ☐

Code First - Answers

1. 260 -> "Out of range". ☐

2. 255 -> "Fits". ☐

3. Unsigned 8-bit values: 2^8 = 256; 0 through 255. ☐

We Do - Check the model

100 is valid upper boundary; 101 is invalid.

Exit Ticket - Check

1. FALSE 2. TRUE 3. FALSE

Explain one false statement using the key ideas.

You Do - Self-mark

Q1 Check input against rules [1]; reject unsuitable data [1]. ☐

Q2 0 and 100: valid boundaries [2]; 101 or -1: invalid [1]. ☐

Q3 Authentication checks identity [1]; validation checks data [1]; iterative tests find faults early [2]. ☐

Try the question again

1. Copy the FULL question you got wrong (or found hardest):

2. In GREEN PEN, write a NEW answer. Include working if needed: